Matgeo Presentation - Problem 9.2.16

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Problem Statement

Find the area of the region bounded by $y = \sqrt{x}$ and $y = x$.

Solution: Matrix and Parametric Forms

The given curves are $y^2 = x$ (for $y \geq 0$) and $y = x$. The matrix form of the parabola $y^2 - x = 0$ is $\mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$, with parameters:

$$\mathbf{V} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} -1/2 \\ 0 \end{pmatrix}, \quad f = 0. \tag{1}$$

The parametric form of the line $y = x$ is $\mathbf{x} = \mathbf{h} + \kappa\mathbf{m}$, with parameters:

$$\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{2}$$

Solution: Intersection Formula

The intersection parameter κ :

$$\kappa_i = \frac{-\mathbf{m}^\top(\mathbf{V}\mathbf{h}+\mathbf{u}) \pm \sqrt{[\mathbf{m}^\top(\mathbf{V}\mathbf{h}+\mathbf{u})]^2 - g(\mathbf{h})(\mathbf{m}^\top\mathbf{V}\mathbf{m})}}{\mathbf{m}^\top\mathbf{V}\mathbf{m}} \tag{3}$$

where $g(\mathbf{h}) = \mathbf{h}^\top\mathbf{V}\mathbf{h} + 2\mathbf{u}^\top\mathbf{h} + f$. With $\mathbf{h} = \mathbf{0}$, we have $g(\mathbf{h}) = 0$ and $\mathbf{V}\mathbf{h} = \mathbf{0}$.

Solution: Calculating Terms

The terms for the formula are:

$$\mathbf{m}^\top \mathbf{V}\mathbf{m} = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \tag{4}$$

$$\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u}) = \mathbf{m}^\top \mathbf{u} = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} -1/2 \\ 0 \end{pmatrix} = -1/2 \tag{5}$$

Substituting into equation (3):

$$\kappa_i = \frac{-(-1/2) \pm \sqrt{(-1/2)^2 - 0}}{1} = \frac{1}{2} \pm \frac{1}{2} \tag{6}$$

Solution: Finding Intersection Points

The values for κ are:

$$\kappa_1 = 1, \quad \kappa_2 = 0 \tag{7}$$

The points of intersection are found by substituting κ_i into the line equation:

$$\mathbf{x_1} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + 1 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{8}$$

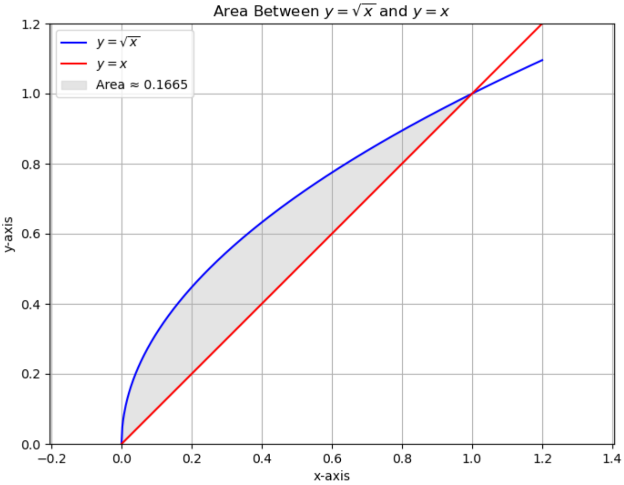
$$\mathbf{x_2} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + 0 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{9}$$

Solution: Area Calculation

The area is given by the integral:

$$\begin{aligned} A &= \int_0^1 (\sqrt{x} - x)\, dx && (10) \\ &= \left[\frac{2}{3}x^{3/2} - \frac{1}{2}x^2\right]_0^1 && (11) \\ &= \left(\frac{2}{3} - \frac{1}{2}\right) - (0) && (12) \\ &= \frac{1}{6} && (13) \end{aligned}$$

Plot


Area Between $y=\sqrt{x}$ and $y=x$
$y=\sqrt{x}$
$y=x$
Area ≈ 0.1665
1.2
1.0
0.8
0.6
0.4
0.2
0.0
y-axis
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
x-axis

C Code: area_calculator.c

```
#include <math.h>

double calculate_area(int n) {
    double a = 0.0;
    double b = 1.0;
    double h = (b - a) / n;
    double sum = 0.0;

    for (int i = 1; i < n; i++) {
        double x_i = a + i * h;
        sum += sqrt(x_i) - x_i;
    }

    double area = h * sum;
    return area;
}
```

Python Code: main_caller.py

```
import ctypes
import os

lib_path = os.path.join(os.getcwd(), ’libarea.so’)
area_calculator_lib = ctypes.CDLL(lib_path)

area_calculator_lib.calculate_area.argtypes = [ctypes.c_int]
area_calculator_lib.calculate_area.restype = ctypes.c_double

num_intervals = 100000
area = area_calculator_lib.calculate_area(num_intervals)

print(f"Calculating␣area␣using␣the␣C␣library...")
print(f"The␣calculated␣area␣is:␣{area}")
print(f"The␣exact␣area␣is␣1/6␣A␣{1/6}")
```

Python Code: plotter.py

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 1.2, 500)
y1 = np.sqrt(x)
y2 = x

fill_x = np.linspace(0, 1, 100)
fill_y1 = np.sqrt(fill_x)
fill_y2 = fill_x

area = np.trapz(fill_y1 - fill_y2, fill_x)

plt.figure(figsize=(8, 6))
plt.plot(x, y1, label=r’$y␣=␣\sqrt{x}$’, color=’blue’)
plt.plot(x, y2, label=r’$y␣=␣x$’, color=’red’)

plt.fill_between(fill_x, fill_y1, fill_y2, color=’lightgray’,
    alpha=0.6, label=f’Area␣A␣{area:.4f}’)
```